

복사전달 특강

Special Topics in Radiative Transfer

Lecture 11
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Based on: Chapter 9 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

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Chapter 9: Extracting Information from MCRT Simulations

MC noise, direct counting, volume-based estimators, and biasing

Individual packet trajectories are not of primary interest — we need **ensemble averages**

Extracting accurate physical quantities requires dealing with the inherent **MC noise**

- Three main strategies presented in this chapter:

9.1 MC Noise

- Fundamental stochastic property
- $\sigma \propto N^{-1/2}$ (central limit theorem)
- Motivates all extraction techniques

9.2–9.3 Estimators

- Direct counting: simple but noisy
- Volume-based estimators (Lucy 1999a): use full trajectory history
- Much less noise for same N

9.4 Biasing

- Importance sampling: invest effort where it matters
- Biased emission, forced scattering, peel-off
- Russian Roulette + composite biasing

9.1 Monte Carlo Noise

The fundamental stochastic fluctuation inherent to all MC calculations

MCRT simulations are **probabilistic**: any quantity extracted will carry stochastic fluctuations

This is called **MC noise** — it is unavoidable but can be reduced by increasing N or using smarter estimators

- The law of large numbers guarantees convergence; the central limit theorem quantifies the noise

MC Estimator (Eq. 90): $G_N = (1/N) \sum_i X_i \rightarrow \text{converges to } E[X] \text{ as } N \rightarrow \infty$

Standard error (Eq. 91): $\sigma_G = \sigma_X / \sqrt{N}$

→ To halve the noise, one must use FOUR TIMES as many packets!

Implications for MCRT Design

- Simple estimates require enormous N for high accuracy
- Noise reduction is a primary driver of algorithm development
- Techniques should maximise the number of useful contributions
- A 'useful contribution' is one that informs the quantity of interest

Example: Escape Probability from a Sphere

- Experiment: packet escapes ($X=1$) or not ($X=0$)
- $G_N \rightarrow$ escape probability with N packets
- $\sigma_G = \sigma_X / \sqrt{N}$ (Fig. 11 demonstrates this)
- High-accuracy result needs $N \sim 10^5 - 10^6$

9.1 MC Noise Scaling: $\sigma \propto N^{-1/2}$

Fig. 11 — Standard error vs number of MC packets

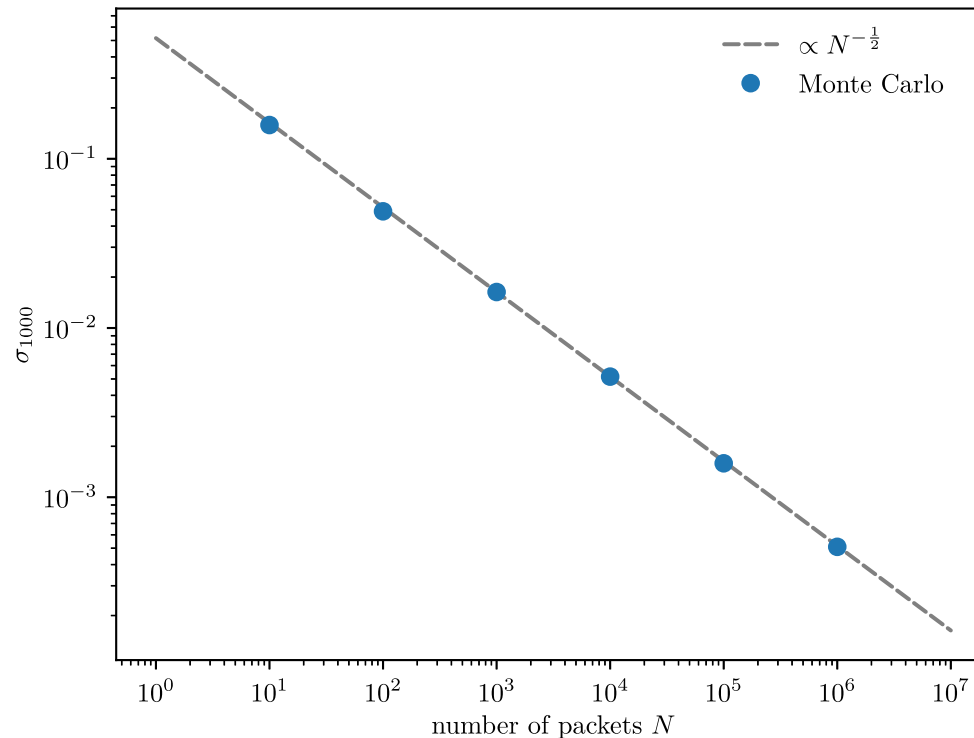


Fig. 11: Standard error σ_{1000} (estimated from 1000 repeated experiments) for the escape probability from a homogeneous sphere ($\tau_{sp} = 1$) as a function of the number of packets N . The result follows $\sigma \propto N^{-1/2}$ (dashed line) almost perfectly, confirming the central limit theorem behaviour.

Key takeaway: accuracy improves slowly with N ($\sigma \propto N^{-1/2}$).

Volume-based estimators and biasing achieve much better accuracy for the same cost.

9.2 Direct Counting of Packets

The simplest — and noisiest — reconstruction approach

Direct counting: query packet positions or properties at discrete instants, then bin/count

- Most natural approach: simple to implement, intuitive, and sometimes sufficient
- Limitation: relies on packets being at the right place at the right time

Synthetic Spectrum / Image (Eq. 92)

$$L(\Delta S_i, \Delta \Omega_j) = (1/\Delta t) \sum_k \epsilon_k$$

- Sum over packets escaping through surface element ΔS_i into solid angle element $\Delta \Omega_j$ in time Δt
- Natural for synthetic spectra and images

Internal Energy Density (Eq. 93)

$$E_i = (1/\Delta V_i) \sum_j \epsilon_j$$

- Sum over all packets inside grid cell i at time t^n
- Straightforward for time-dependent simulations
- Requires many packets in each cell $\rightarrow$ can be very noisy

For time-independent problems: requires packets to be **in the right cell at the right time** — rare event!

- For optically thin regions: few packets interact $\rightarrow$ high noise

Still useful as a **reference/validation** for more sophisticated estimators

9.3 Volume-Based Estimators — Concept

Lucy (1999a): use trajectory information instead of instantaneous snapshots

Key idea: instead of snapshots, use the **full trajectory** of each packet through each cell

- A packet that traverses a cell contributes information to ALL cells along its path — not just one
- This dramatically increases the number of contributions per cell → much lower noise

Volume-Based Estimator principle (Lucy 1999a):

Each trajectory segment j of length l_j (in cell ΔV_i) contributes to the estimator weighted by $l_j/c\Delta t$.

→ Both forward-scattered AND backscattered contributions are included!

Advantages Over Direct Counting

- A single packet contributes to multiple cells
- Backscattered packets also contribute (can re-enter a cell)
- Non-zero result as long as ANY packet passed through the cell
- Especially powerful in optically thin regions

Physical Basis

- The specific intensity is an integral over angle
- Packet trajectory elements statistically represent the radiation field
- Time-averaged properties are more stable than snapshots
- Applicable to both time-dependent and steady-state RT

9.3.1 Example: Energy Density Estimator

Deriving the volume-based estimator for the radiation energy density

- Consider a packet with energy ε propagating in cell ΔV_i during simulation time step Δt
- The packet spends time δt_j on segment j ; its contribution to energy density $\sim \delta t_j / \Delta t \cdot \varepsilon_j$

Time-based estimator (Eq. 94): $E_i = (1/\Delta V_i) \sum_{j \in \Delta V_i} (\delta t_j / \Delta t) \varepsilon_j$

Since packets travel at speed c : $l_j = c \delta t_j$

→ Length-based estimator (Eq. 95): $E_i = (1/\Delta t c \Delta V_i) \sum_{j \in \Delta V_i} l_j \varepsilon_j$

Key Properties

- Sum includes ALL trajectory segments in cell ΔV_i (physical scatterings AND numerical events like cell crossings)
- The ratio $\varepsilon_j / \Delta t$ is independent of the absolute time interval
- Applicable to time-independent steady-state problems!

Practical Implementation

- During propagation loop: at each segment, add $l_j \varepsilon_j$ to cell estimator
- After simulation: divide by $\Delta t c \Delta V_i$
- For time-independent: Δt cancels out (only ratios matter)
- Easily extended to other radiation field quantities

9.3.2 Constructing Volume-Based Estimators: Radiation Field Quantities

Estimators for mean intensity J , flux H (or F), and second moment K

Using $E = (4\pi/c)J$ [Eq. 96] and the VBE for E [Eq. 95]:

Mean Intensity J_i [Eq. 97]

$$J_i = (1/4\pi \Delta V_i \Delta t) \sum_{j \in \Delta V_i} l_j \epsilon_j$$

- Restrict sum to segments in direction $\Delta\Omega_k \rightarrow$ specific intensity $I_{j,k}$ [Eq. 98]
- Further restrict by frequency $\rightarrow$ monochromatic $I_{j,k,p}$ [Eq. 99]

Radiation Flux F_i [Eq. 100]

$$F_i = (1/\Delta V_i \Delta t) \sum_{j \in \Delta V_i} n_j l_j \epsilon_j$$

- n_j : propagation direction of segment j
- Similarly: K moment includes $n^i n^j$ weighting
- All moments easily obtained by including powers of direction cosines

- VBE for J : same packet contributes to multiple cells along its path (not just final cell)
- Demonstrated with homogeneous sphere test — Fig. 12: VBE recovers J , H , K to high accuracy
- Python implementation: `update_estimators()` routine in `mcrt_hom_sphere.py` (Appendix B)

9.3.2 Test: Volume-Based Estimators for J, H, K

Fig. 12 — Homogeneous sphere test (Abdikamalov et al. 2012 parameters)

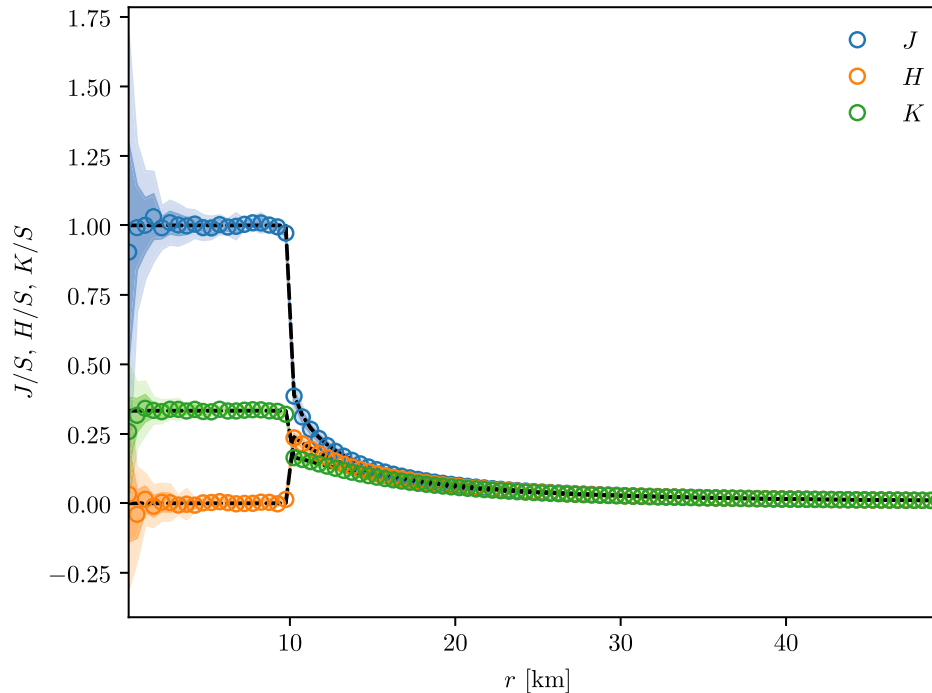


Fig. 12: Volume-based estimators for J (blue), H (orange), K (green) in the homogeneous sphere test. Shown relative to the source function S. Shaded bands: 1 σ and 2 σ confidence from 10 MCRT runs (10⁵ packets each). Open symbols: mean values. Black dashed: analytic solution [Eqs. A.5–A.10]. Agreement is excellent throughout, demonstrating the power of the VBE approach.

9.3.2 Direct Counting vs Volume-Based Estimator

Fig. 13 — Why VBE dramatically outperforms direct counting

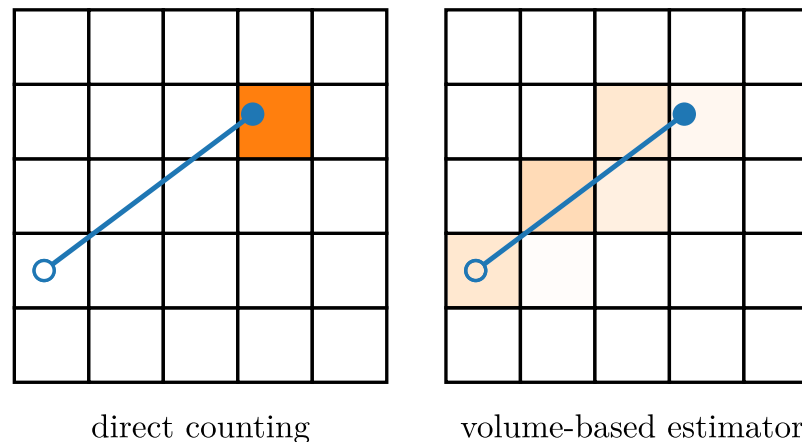


Fig. 13: Single packet trajectory (blue) absorbed at the filled circle. LEFT (direct counting): packet contributes heating rate only in the FINAL cell where absorbed. RIGHT (volume-based estimator): packet contributes to the heating rate estimator in ALL cells it traversed — weighted by the optical depth of each segment. Color transparency encodes the relative contribution to each cell.

Direct Counting Problem

- Only one cell gets information per packet
- All other traversed cells: wasted effort
- Optically thin: rarely absorbed $\rightarrow$ high noise

VBE Solution

- All traversed cells benefit from trajectory
- Non-zero result even if none absorbed
- Better statistics $\rightarrow$ lower noise for same N

9.3.3 Constructing Volume-Based Estimators: Extracting Physical Rates

Heating rates, photoionization rates, and momentum deposition

Volume-based approach extends naturally to **physical rates** — not just radiation field moments

- General principle: any rate that depends on the radiation field → substitute VBE for the field
- The optical depth of each segment weights how much interaction the packet COULD have experienced

Heating rate estimator (Eq. 103): $\Delta \dot{E}_i = (1/\Delta V_i \ c \Delta t) \sum_{j \in \Delta V_i} \chi \ l_j \ \epsilon_j$

- χ : absorption coefficient (not scattering)
- Gives the total rate of energy absorbed from the radiation field in cell i

Generalisation

- Any rate Q proportional to $\int dv \ \chi_v \ J_v$ can be estimated:
 $\hat{Q}_i = (1/\Delta V_i \ c \Delta t) \sum_j \chi_v(v_j) \ l_j \ \epsilon_j$
- The sum weights each segment by the appropriate cross-section
- Non-zero as long as ONE packet passed through the cell

Important Note: Line Interactions

- For Sobolev-based line interactions, estimators are sums over resonance events (not trajectory segments)
- All resonances contribute even without an actual interaction
- Critical for line-driven wind calculations (Noebauer & Sim 2015)

9.3.4 Example: Photoionization Rate Estimators

A concrete illustration of the VBE principle for atomic rates

- Photoionization rate coefficient $\gamma = 4\pi \int_{\nu_{\text{th}}}^{\infty} (\sigma_{\nu}/h\nu) J_{\nu} d\nu$
- σ_{ν} : photoionization cross section; ν_{th} : threshold frequency
- Substitute the VBE for J_{ν} [Eq. 97] $\rightarrow$ direct estimator for γ

Photoionization rate estimator (Eq. 102): $\hat{\gamma} = (1/\Delta V_i \Delta t) \sum_{j \in \Delta V_i, \nu > \nu_{\text{th}}} (\sigma(\nu_j)/h\nu_j) I_j \epsilon_j$

- Sum restricted to segments with $\nu > \nu_{\text{th}}$ (photon above threshold)
- Each segment weighted by photoionization cross-section at that frequency

Significance

- Result is non-zero as long as SOME packet above ν_{th} passed through the cell
- Even one packet gives a statistically useful estimate
- Compare: direct counting requires an actual photoionization event in that cell
- Especially useful in optically thin, highly ionized regions

9.3.5 Energy and Momentum Flow

Similar VBE for radiation flux (Eq. 100):

$$F_i = (1/\Delta V_i \Delta t) \sum_j n_j I_j \epsilon_j$$

- Used for radiation momentum deposition in RH calculations
- Particularly important for radiation-driven outflows (Sect. 11)
- Instrument for coupling RT to fluid dynamics

9.4 Biasing — Importance Sampling

Selectively investing computational effort where it matters most

In many MCRT applications, only a **subset** of packet trajectories contributes to the observable of interest

- Example: synthetic image — only packets escaping toward the observer are relevant

Biasing (importance sampling): modify the sampling PDF to over-represent relevant packets

Biasing introduces a new PDF $q(x) \neq p(x)$. To maintain physical consistency:

Adjust packet weight: $w(x) = p(x)/q(x) \cdot w_{nb}$ [Eq. 104]

→ Packets from over-represented regions carry lower weight; under-represented regions carry higher weight.

Key Principle

- Biasing increases statistics where relevant
- Complementary decrease in statistics elsewhere
- Only useful if the 'elsewhere' packets would not contribute anyway
- Widely used in dust RT: Steinacker et al. (2013)

Biasing Techniques Covered (Sect. 9.4)

- 9.4.1 Biased emission
- 9.4.2 Forced scattering
- 9.4.3 Peel-off (next event estimate)
- 9.4.4 Further techniques
- 9.4.5 Russian Roulette & composite biasing

9.4.1 Biased Emission

Over-representing emission from weak or distant sources

- Problem: two sources with very different luminosities $L_1 \ll L_2$
- Unbiased: N packets with equal energy $\epsilon = (L_1 + L_2)\Delta t/N$ — source 1 represented by very few packets
- Biased: equal number of packets from each source, but with adjusted weights

Unbiased Approach

- Probability of packet from source i : $p_i = L_i/L$ [Eq. 106]
- For $L_1 \ll L_2$: source 1 represented by $\ll N/2$ packets
- High noise for source 1 properties
- Each packet energy: $w = \epsilon = L\Delta t/N$ [Eq. 105]

Biased Approach

- Use uniform $p_1 = p_2 = 1/2$ [Eq. 107]
- Adjust weights: $w_1 = 2(L_1/L)\epsilon$ [Eq. 108]
 $w_2 = 2(L_2/L)\epsilon$ [Eq. 109]
- Both sources equally well-sampled statistically
- Energy conservation maintained via weight adjustment

Also used to preferentially launch packets in **directions of particular interest** (e.g., toward observer)

- Detailed account: Baes et al. (2016) for dust RT applications

9.4.2–9.4.3 Forced Scattering and Peel-off

Two widely-used biasing techniques to improve synthetic observables

9.4.2 Forced Scattering

- Problem: in optically thin media, packets escape without scattering → poor statistics
- Solution: FORCE at least one interaction before domain edge
- Biased PDF (Eq. 110): $q(\tau) d\tau = \exp(-\tau) / (1 - \exp(-\tau_{\text{edge}})) d\tau$ for $\tau \leq \tau_{\text{edge}}$
- Weight adjusted (Eq. 111): $w(\tau) = 1 - \exp(-\tau_{\text{edge}})$
- Warning: continual application → termination via Russian Roulette needed

9.4.3 Peel-off (Next Event Estimate)

- Problem: few packets escape toward a specific observer direction
- Peel-off: at every interaction, compute probability that packet COULD have escaped toward observer
- Weight contributed (Eq. 112):
 $w_{\text{obs}} = w \cdot p(n_{\text{obs}}) \cdot \exp(-\tau_{\text{obs}})$
 $p(n_{\text{obs}})$: re-emission probability in observer direction
 $\exp(-\tau_{\text{obs}})$: attenuation along path to observer

Peel-off improvement: every interaction contributes to observer spectrum — not just escaping packets

- Peel-off is particularly powerful for spectral synthesis in SN ejecta and dust RT (Steinacker et al. 2013)
- Drawback: ray-tracing to observer adds computational cost but usually worth it for noise reduction

9.4.4–9.4.5 Further Biasing Techniques and Limitations

Path length stretching, composite biasing, and Russian Roulette

9.4.4 Further Biasing Techniques

- Path length stretching (Baes et al. 2016)
- Continuous absorption / packet splitting / survival biasing
(Carter & Cashwell 1975; Lee et al. 2017)
- Polychromatism (Jonsson 2006; Steinacker et al. 2013)
- See Steinacker et al. (2013) review for comprehensive overview

9.4.5 Limitations of Biasing

- Biasing in one region → LOSS of statistics in others
- High-weight packets dominate MC noise when under-represented
- Weights can become unbounded in principle!
- Composite biasing (Baes et al. 2016): blend biased + original PDF

$$q^*(x) = (1-\zeta)p(x) + \zeta q(x) \rightarrow w^* < 1/(1-\zeta) \text{ [Eq. 115]}$$

Russian Roulette — Terminating Low-Weight Packets

- Very low-weight packets contribute negligibly but cost the same computationally
- Russian Roulette: define termination probability p_T
- Sample: with probability p_T → terminate the packet; with $(1-p_T)$ → survive with weight w/p_T
- Statistically unbiased: surviving packets carry the redistributed weight of terminated ones
- Applied once weight drops below a threshold; details in Dupree & Fraley (2002)

Chapter 9 — Summary

Extracting accurate physical information from MCRT simulations

MC Noise and Estimators

- Fundamental limit: $\sigma \propto N^{-1/2}$
- Direct counting: intuitive but noisy
- Volume-based estimators (Lucy 1999a): use full trajectory, much less noise
- VBE for E, J, H, K, γ , heating rate... all constructable

Biasing Techniques

- Importance sampling: invest effort where it matters
- Biased emission: equal representation of all sources
- Forced scattering: guarantee interactions in thin media
- Peel-off: every interaction contributes to observer
- Russian Roulette: cull low-weight packets safely

- VBE + biasing together allow high-accuracy results with moderate N — essential for practical applications
- These techniques underpin sophisticated codes like ARTIS, SEDONA, TARDIS for SN spectral synthesis

Next: special MC techniques for **optically thick** regimes (Chapter 10 — IMC and diffusion MC)